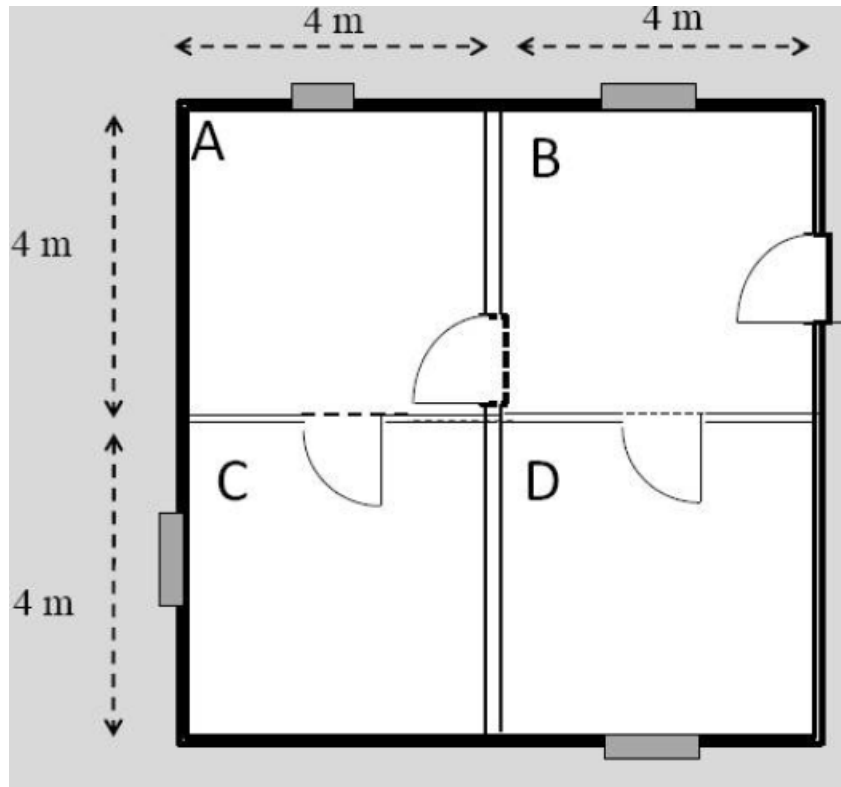


Temperature control in a four-rooms building

Consider the problem of controlling the temperature of the building



whose mathematical model is

$$\begin{aligned} c\rho V\dot{T}_A &= s_r u_2(T_B - T_A) + s_r u_1(T_C - T_A) + s_e U(T_E - T_A) + q_A \\ c\rho V\dot{T}_B &= s_r u_2(T_A - T_B) + s_r u_1(T_D - T_B) + s_e U(T_E - T_B) + q_B \\ c\rho V\dot{T}_C &= s_r u_1(T_A - T_C) + s_r u_2(T_D - T_C) + s_e U(T_E - T_C) + q_C \\ c\rho V\dot{T}_D &= s_r u_1(T_B - T_D) + s_r u_2(T_C - T_D) + s_e U(T_E - T_D) + q_D \end{aligned}$$

c : specific heat of the air; V : volume of the rooms; T_i , $i = A, B, C, D$: temperatures of the rooms; s_r : surface of the walls between two rooms; s_e : surface of the walls between a room and the environment; u_1, u_2, U : transmittances; T_E : external temperature; q_i , $i = A, B, C, D$: inputs (heating power).

- Equilibrium condition: $T_E = 0^\circ C$, $T_A = T_B = T_C = T_D = \bar{T} = 20^\circ C$,
 $q_A = q_B = q_C = q_D = \bar{q} = s_e \cdot U \cdot \bar{T}$.
- We define: $\delta T_A = T_A - \bar{T}$, $\delta T_B = T_B - \bar{T}$, $\delta T_C = T_C - \bar{T}$, $\delta T_D = T_D - \bar{T}$,
 $\delta q_A = (q_A - \bar{q})/c\rho V$, $\delta q_B = (q_B - \bar{q})/c\rho V$, $\delta q_C = (q_C - \bar{q})/c\rho V$ and
 $\delta q_D = (q_D - \bar{q})/c\rho V$.

Around the given equilibrium condition:

$$\begin{bmatrix} \dot{\delta T}_A \\ \dot{\delta T}_B \\ \dot{\delta T}_C \\ \dot{\delta T}_D \end{bmatrix} = \begin{bmatrix} -\gamma_{ol} & \gamma_2 & \gamma_1 & 0 \\ \gamma_2 & -\gamma_{ol} & 0 & \gamma_1 \\ \gamma_1 & 0 & -\gamma_{ol} & \gamma_2 \\ 0 & \gamma_1 & \gamma_2 & -\gamma_{ol} \end{bmatrix} \begin{bmatrix} \delta T_A \\ \delta T_B \\ \delta T_C \\ \delta T_D \end{bmatrix} + \begin{bmatrix} \delta q_A \\ \delta q_B \\ \delta q_C \\ \delta q_D \end{bmatrix}$$

where $\gamma_1 = \frac{s_r u_1}{c\rho V}$ and $\gamma_2 = \frac{s_r u_2}{c\rho V_A}$, $\gamma_e = \frac{s_e U}{c\rho V}$, $\gamma_{ol} = \gamma_1 + \gamma_2 + \gamma_e$.
 Importantly, $\gamma_{ol} > \gamma_1 > \gamma_2$.

The system “centralized” model

$$\dot{x} = Ax + Bu$$

is given in the corresponding MATLAB file, where $x = [\delta T_A, \delta T_B, \delta T_C, \delta T_D]^T$ and $u = [\delta q_A, \delta q_B, \delta q_C, \delta q_D]$. We assume that all the state variables are measurable.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time selected compatibly with the continuous-time dynamics). Perform the following analysis:

- a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
- b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
- 3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop trajectories (generated both in continuous-time and in discrete-time) of the room temperatures starting from a common random initial condition.